

5(a)(i)	<i>four from:</i> 1 obtain (dominant <i>APP</i>) allele ; 2 detail ; <i>any one from:</i> - e.g. synthesise gene - make cDNA from mRNA - use probe - select and amplify with PCR - gel electrophoresis 3 restriction enzyme ; 4 use, vector / plasmid / virus ; A gene gun / direct (micro)injection 5 (on) zygote / secondary oocyte / egg (cell) / early embryo ; I sperm 6 AVP ; e.g. cloning / embryo splitting - add promoter - marker gene / tag gene	4
5(a)(ii)	<i>one from:</i> to test treatments without harming humans ; to investigate, cause / progress, of disease ;	1
5(b)	<i>four from:</i> 1 identifies, active / switched on / expressed / transcribed, genes ; 2 transcription of a gene produces mRNA ; 3 ssDNA act as, probes / reporters ; 4 (ssDNA) bound at known positions to a, solid surface / slide / chip ; 5 cDNA, binds to / hybridises with, complementary (probe) ssDNA ; 6 show up / identified as, fluorescent spots / named colour ; 7 positions / intensity, recorded by, laser / scanner ; 8 positions identified as named genes ; 9 intensity proportional to gene expression ;	4

Question	Answer	Marks
5(c)(i)	2.7 (%) ; ; A 2.68 allow 1 working mark for: $932 - 28 = (904)$ $904 \div 33\ 696$	2
5(c)(ii)	training causes more genes to be expressed ;	1
5(d)(i)	gene therapy ;	1
5(d)(ii)	<i>two from:</i> 1 switches on genes needed for forming, memories / synapses ; 2 allows better learning (of water maze) ; 3 reduces, memory loss / symptoms of Alzheimer's ;	2

Question	Answer	Marks
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